

C2DB

16,905 entries
PBE / HSE06 / G_0W_0
masses, polarizability

JARVIS-DFT 2D

1,103 monolayers
131 structure matches
 σ for ϵ

Screening

stability, gap > 0.3 eV
non-magnetic
parabolic bands
 $\rightarrow$ 1085 materials

Gap calibration

$E_g \sim a + b_1 E_{\text{PBE}} + b_2 E_{\text{HSE}}$
129 G_0W_0 anchors
10-fold coverage check

Uncertainty

E_g : measured
 ϵ : measured
 m^* : ASSUMED

Device model

ballistic, ambipolar
double gate
 $L_g = 12$ nm
 $\rightarrow$ 754 materials

Monte Carlo

10^3 draws per material

Rank stability

dominance, Kendall τ , entropy

Sensitivity

Sobol; E_g acts in 70/754

Robust rule

10th-percentile lower bound